Unit 3: Equations of Lines and Planes

Complete the following table:

Type of Equation	Lines in R^2	Lines in R^3	Planes (in R^3)
Vector			
Parametric			
Symmetric			
	** can have combination of parametric and symmetric if either $a, b, c = 0$		
Cartesian (Scalar)			

	Parallel if:	Perpendicular if:
Lines are		
Planes are		

	Line in R^2	Line in R^3	Plane in R^3
Minimum Distance from a point to	$d =$	$d =$	$d =$

	Long way to check:	Fast way to check:
Three vectors in R^3, $\vec{u}, \vec{v}, \vec{w}$ are coplanar		

Practice Questions

1. In the vector equation $\vec{r} = (3,2,-1) + t(4,0,-2),\ t \in R$, a point on the line is _____ and a direction vector for the line is _____.

2. In the parametric equations $x = -3 + 2t,\ y = 4 + t,\ z = 2 + 7t,\ t \in R$, a point on the line is _____, and a direction vector for the line is _____.

3. In the symmetric equations $\frac{x-1}{3} = \frac{2-y}{4} = z + 5$, a point on the line is _____ and a direction vector of the line is _____.

4. The minimum distance of the point $Q(2,-4)$ to the line $2x + 3y - 7 = 0$ is _____.

5. In a vector equation, $\vec{r} = (3,-7,2) + s(-2,4,1) + t(-1,1,1),\ s, t \in R$, a point on the plane is _____ and two direction vectors for the plane are _____ and _____.

6. Vectors $\vec{u}$ and $\vec{v}$ are a basis for a plane provided they are _____.

7. The plane $2x - 5y + 3z - 4 = 0$ has a normal vector $\vec{n}$ = _____ .

8. If $\vec{u}, \vec{v},$ and $\vec{w}$ are three vectors lying in the same plane and if $\vec{u}$ and $\vec{v}$ are non- collinear, then $\vec{w}$ = _____ . These vectors are said to be _____.

9. If $P(3,-7,4), Q(1,3,6)$ and $R(-1,2,-5)$ are three points in a plane, then two vectors in the plane are _____ and _____.

10. To find the normal to the plane in #9, you should find _____.

11. The Cartesian equation of the plane $\vec{r} = (-2,3,2) + s(1,-3,0) + t(-3,1,1),\ s, t \in R$ is _____.

12. The planes $2x - 5y + 4z + 12 = 0$ and $4x - 10y + 8z = 1$ are parallel because _____.

13. To quickly determine if three vectors form a basis for space, what calculation should be performed? _____.

ANSWERS:

1. point: $(3,2,-1)$; direction vector: $(4,0,-2)$ (or any scalar multiple!)
2. point: $(-3,4,2)$; direction vector: $(2,1,7)$ (or any scalar multiple!)
3. point: $(1,2,-5)$; direction vector: $(3,-4,1)$ (or any scalar multiple!)
4. $\frac{15\sqrt{13}}{13}$ units
5. point: $(3,-7,2)$; direction vectors: $(-2,4,1), (-1,1,1)$ (or any scalar multiples of those!)
6. not collinear (i.e. linearly independent)
7. $(2,-5,3)$
8. $\vec{w} = a\vec{u} + b\vec{v}$; coplanar
9. $(-2,10,2)$ or $(2,1,11)$ or $(4,-9,9)$ (many answers possible – create a vector between any pair of points and/or use scalar multiples of any of those vectors created)
10. the cross product of any two non-collinear direction vectors
11. $3x + y + 8z - 13 = 0$
12. their normal vectors are scalar multiples of one another
13. the triple scalar product